Supervized learning :

Sınıflandırma:

$$\mathcal{X} = \{x^t, r^t\}_{t=1}^{N}$$ (r^t: etiket)

$x^t \in C^i \Rightarrow$ sınıflandırma

$$h_i(x^t) = \begin{cases} 1 & \text{if } x^t \in C^i \\ 0 & \text{if } x^t \in C_j, j \neq i \end{cases}$$

$$E\{\mathcal{E}_n \mid X\} = \sum_{t=1}^{N} \sum_{i=1}^{N} 1_{h_i(x^t) \neq r_i^t}$$

($E\{\mathcal{E}_n \mid X\}$: sınıflandırmadaki hata)

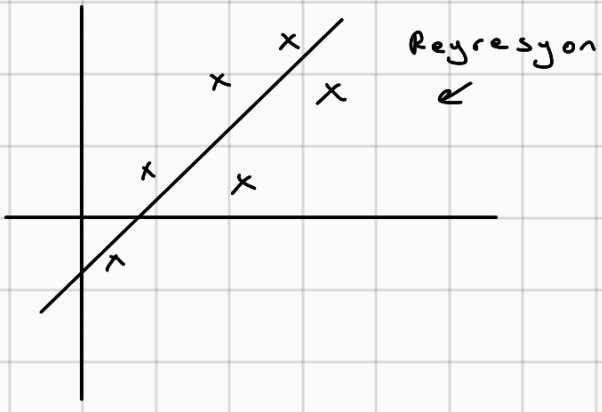
Regresyon:

$r^t \in \mathbb{R}$

$\hat{r}^t = f(x^t)$ (kestirim)

$r^t = \hat{r}^t = f(x^t) \Rightarrow$ interpolasyon

$\hat{r}^t = f(x^t) + \underbrace{\varepsilon}_{\text{gürültü}}$ (regresyon)



$$E\{\mathcal{E}_g \mid X\} = \frac{1}{N} \sum_{i=1}^{N} [r^t - g(x^t)]^2$$

(E: Expected value)

Loss f.

$r^t = f(x^t) + \underbrace{\varepsilon}_{\text{gürültü}}$ fonksiyonun

$g(x)$ ile tahmin etmeye çalışıyoruz.

En basit modellene : doğrusal modelleme

$$g(x) = w_1 x_1 + w_2 \cdot x_2 + \text{---} + w_N x_N + w_0$$

$$= \vec{w} \cdot \vec{x} + w_0$$

$$g(x) = w_1 x_1 + w_0$$

$$E\{\underbrace{w_1, w_0}_{\text{değişkenleriniz bunlar}} | X\} = \frac{1}{N} \sum_{i=1}^{N} \left(r^t - (w_1 x_1 + w_0) \right)^2$$

$$\frac{\partial E\{w_1, w_0 | X\}}{\partial w_0} = -\frac{1}{N} \sum_{i=1}^{N} 2\left(r^t - (w_1 x_1 + x_0)\right)$$

$$\frac{\partial E\{w_1, w_0 | X\}}{\partial w_1} = \frac{1}{N} \sum_{i=1}^{N} 2\left[r^t - (w_1 x_1 + w_0)\right](-w_1) = 0$$

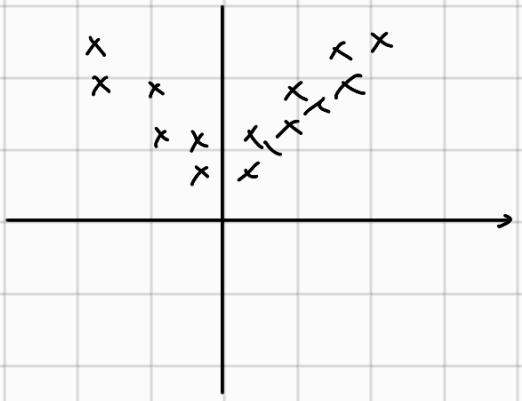
$$w_1 = \frac{\sum_{t=1}^{N} x^t \cdot r^t - \bar{x}\,\bar{r}\,N}{\sum_{t=1}^{N} (x^t)^2 - N \cdot \bar{x}^2}$$

$$\bar{r} = \frac{1}{N} \sum_{t=1}^{N} r^t$$

$$\bar{x} = \frac{1}{N} \sum_{t=1}^{N} x^t$$

$$w_0 = \bar{r} - w_1 \bar{x}$$

ya böyle olsaydı



$$g(x) = w_2 x_1^2 + w_1 x + w_0$$

ikinci bir boyut ekledik $[x_1^2, x_1]$

2, 3
1, 4

$[4, 2] \rightarrow 3$
$[1, 1] \rightarrow 4$

$\{x^t, r^t\}_{t=1}^{N}$

$g(x) = w_1 x_1 + w_0$

3 adımımız var

① Model oluşturma

② Maliyet fonksiyonu

$$E[\theta|x] = \sum_t L(r^t, g(x^t|\theta))$$

③ Optimizasyon Yöntemi

Hastalık tespitinde

$$\sum_{k=1}^{K} P(C_k|x) = 1$$

$$R\{\alpha_i|x\} = \sum_{k=1}^{K} \lambda_{ik} \cdot P(C_k|x)$$

λ_{ik}: risk fonksiyonu

C_1 : sağlıklı

C_2 : kanser

λ_{ij}	C_1	C_2
α_1	0	1000
α_2	10	100

0/1 loss: $\lambda_{ik} = \begin{cases} 0 & \text{if } i = k \\ 1 & \text{o.w} \end{cases}$

$$R\{\alpha_i|x\} = \sum_{k=1}^{K} \lambda_{ik} \cdot P(C_k|x)$$

$$= \sum_{k \neq i} P(C_k|x)$$

$$= 1 - P(C_i|x)$$

$h_i(x^t)$ aidat fonksiyonu

C_i s.t $\underset{i}{\operatorname{argmax}}\ h_i(x^t)$

Çocuk bezi , tıraş bıçağı

$$\frac{P(x,y)}{P(x)\cdot P(y)}$$

beraberalınmış

$$\frac{100}{100+50+150} = \frac{1}{3}$$

x ile y arasındaki ilişkilerine bakılır

İlişki kuralları

(1) $P(x,y) =$

(2) $P(Y|X) = \frac{P(x,Y)}{P(x)}$ X'leri satın almasının kaçında yleri de satın alınmış

(3) $\frac{P(Y|X)}{P(Y)} = \frac{P(x,y)}{P(x)P(Y)}$ tamamen bağımsızlık $P(x,y) = P(x).P(y)$ olur

Korelasyon ile nedensellik aynı şey değil

Bayes kural

$P(z|\theta)$ — θ: parametre vektörü; bunları bulma çalışıyoruz

$P(x;\theta)$ — θ kümesi için X kümesinin olasılığı

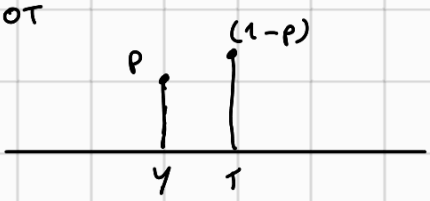
$x^t \sim P(x|\theta)$ → theta parametresi ile gösterilen P dağılımına uygun

$$\theta_{MLE} = \arg\max_{\theta} P(z|\theta)$$

$$l(\theta|X) = P(X|\theta) = \prod_{t=1}^{N} P(x^t|\theta)$$

Yazı tura

SOY
GOT

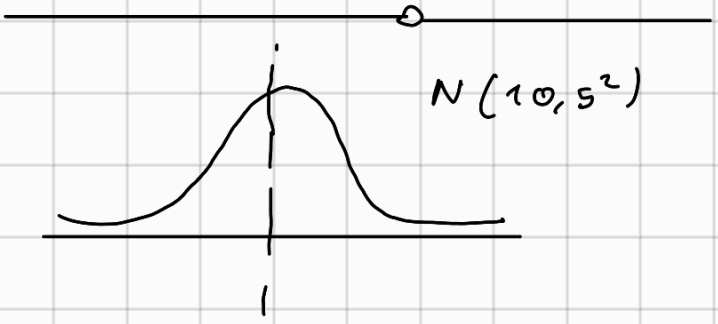


$$l(\theta|X) = P(X|\theta) = \prod_{t=1}^{N} p(x^t|\theta)$$

↓ çarpımdan kurtulmak için

$$L(\theta|X) = \log\left(\sum_{t=1}^{N} p(x^t|\theta)\right)$$

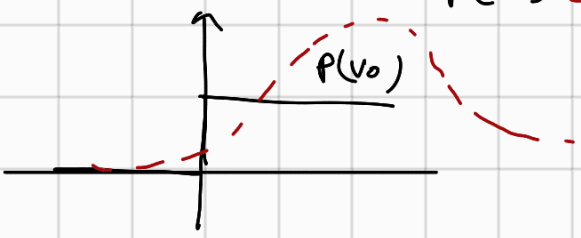
hemde üstel olasılık dağılımları çok fazlı



Bayesian Theory :

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

posterior ← $P(X|\theta) = \frac{P(\theta|X)\ P(X)}{P(\theta)}$ ← likelihood, prior, evidence



$$\hat{\theta} = \underset{a}{\operatorname{argmin}}\ E\{L(\theta, a)|X\}$$

↓ karar

MMSE (Minimum mean square error).

$$C(\theta, a) = \|\theta - a\|_2^2$$

$$\bar{\theta} = E[\theta | X]$$

$$E\{C(\theta, a) | X\} = E\{(\theta - a)^T (\theta - a) | X\}$$

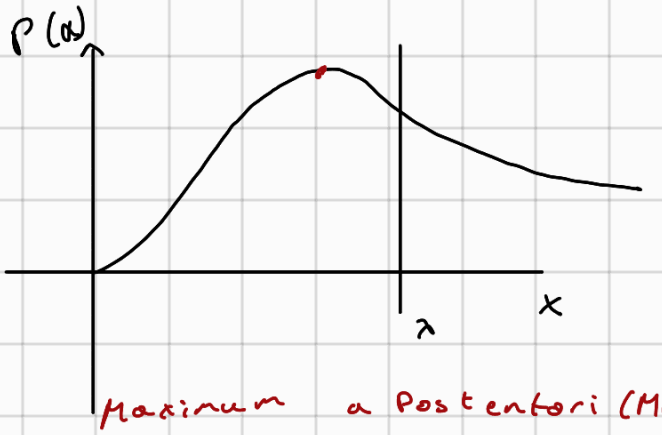
$$\frac{\partial}{\partial a} E\{(\theta - a)^T (\theta - a) | X\} = 0$$

$$\frac{\partial}{\partial a} E\{\theta\theta^T - \theta \cdot a^T - a\,\theta^T + a\,a^T | X\} = 0$$

$$0 - \frac{\partial \bar{\theta} a^T}{\partial a} - \frac{\partial a \bar{\theta}^T}{\partial a} + \frac{\partial (a\,a^T)}{\partial a} = 0$$

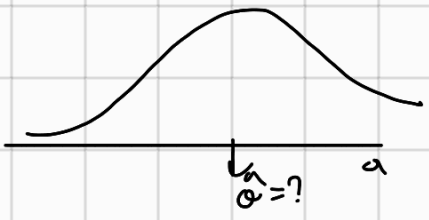
$$-\theta^T - \bar{\theta} + 2a^T = 0 \quad \Rightarrow a = \bar{\theta}$$

$$\hat{\theta} = \bar{\theta}$$



Maximum a Posteriori (MAP) Estimation

$$C(\theta, a) = -\delta(\theta - a)$$



$$E\{C(\theta, a)\} = E\{-\delta(\theta - a) | X\}$$

$$= -\int p(\theta | x)\, \delta(\theta - a)\, d\theta$$

$= -P(\theta|x)|_{\theta=a}$

$\hat{\theta}_{MAP} = \arg\min_{u} \left(-P(\theta|x)|_{\theta=u}\right)$

$\hat{\theta} = \arg_u \max P(\theta|x)$